HDOC- 7 – 1
NAME : DARSHIL RAJ
DATE : 4/10/26 SAP ID: 590040196

Program to Check Whether a Character is a Vowel or Consonant Using if–else

This document provides a clean, well-commented implementation for checking whether a character is a vowel or a consonant in C. The program takes a character as input and uses if–else statements with logical operators to decide and display the result.

1. C Implementation

Step 1: Understand the Logic

- The vowels are a, e, i, o, u (and their uppercase forms A, E, I, O, U).
- Any other English letter is a consonant.
- A character that is not a letter (such as a digit or symbol) is neither a vowel nor a consonant.
- The logical OR operator (||) is used to compare the character with each vowel.

Step 2: Define Variables and Data Types

- Use a character variable (char) to store the input character.

Step 3: Take User Input

- Display a prompt using printf("Enter a character: "), then read the value using scanf(" %c", &ch).
- The space before %c skips any leftover whitespace or newline.

Step 4: Apply the if–else Conditions

1. Outer if: check whether the character is a letter (a–z or A–Z).
2. Inner if: if the character matches any vowel, it is a vowel.
3. Inner else: the character is a consonant.
4. Outer else: the character is not an alphabet.

Step 5: Format the Output

- Print the character along with the message "vowel", "consonant" or "not an alphabet" using printf().

Complete C Code

```c
/*
 * Q9 (User Inputs, Operations & Output)
 * Write a program to input a character and check whether it is
 * a vowel or consonant using if-else.
 */
#include <stdio.h>

int main() {
    char ch;

    // Step 1: Read the character
    printf("Enter a character: ");
    scanf(" %c", &ch);

    // Step 2: Check whether it is an alphabet
    if ((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
        // Step 3: Check for vowels (lowercase and uppercase)
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
            printf("%c is a vowel.\n", ch);
        } else {
            printf("%c is a consonant.\n", ch);
        }
    } else {
        printf("%c is not an alphabet.\n", ch);
    }

    return 0;
}
```

Program Output (Sample Runs)

Run 1:

```
$ gcc hdoc7_1.c -o hdoc7_1
$ ./hdoc7_1
Enter a character: e
e is a vowel.
```

Run 2:

```
$ ./hdoc7_1
Enter a character: K
K is a consonant.
```

Run 3:

```
$ ./hdoc7_1
Enter a character: 5
5 is not an alphabet.
```

Dry Run with Test Cases

Sample Test Case 1:

- Input: e
- Calculation:
  - Outer if: 'e' lies between 'a' and 'z', so it is a letter (TRUE)
  - Inner if: ch == 'e' is TRUE, so the vowel block runs
- Output: e is a vowel.

Sample Test Case 2:

- Input: K
- Calculation:
  - Outer if: 'K' lies between 'A' and 'Z', so it is a letter (TRUE)
  - Inner if: 'K' matches none of a, e, i, o, u, A, E, I, O, U (FALSE)
  - The inner else runs
- Output: K is a consonant.

Sample Test Case 3:

- Input: 5
- Calculation:
  - Outer if: '5' is not between 'a'–'z' or 'A'–'Z' (FALSE)
  - The outer else runs
- Output: 5 is not an alphabet.